MIL-H-12394D
17 June 1976
SUPERSEDING
MIL-H-12394C
6 February 1968

MILITARY SPECIFICATION

HEATERS, WATER, LIQUID FUEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers direct fired, oil-burning, forced-draft, hot-water-supply boilers (hereinafter referred to as "water heaters"), designed for installation with separate hot water storage tanks.

1.2 Classification. The water heaters shall be of the following sizes, as specified (see 6.2).

Size 600 - 600 gallons per hour at 50°F temperature rise (10 gpm).
Size 2,000 - 2,000 gallons per hour at 50°F temperature rise (33-1/2 gpm).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

J-C-30	- Cable and Wire, Electrical (Power, Fixed Insulation).
W-F-406	- Fittings for Cable, Power, Electrical and Conduit, Metal, Flexible.
QQ-I-652	- Iron Castings, Gray.

FSC 4520

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QQ-S-775	- Steel Sheets, Carbon, Zinc-Coated.
VV-F-800	- Fuel Oil, Diesel.
VV-F-815	- Fuel Oil, Burner.
WW-C-566	- Conduit, Metal, Flexible.
WW-C-581	- Conduit, Metal, Rigid; and Coupling, Elbow, and Nipple, Electrical Conduit: Zinc-Coated.

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MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-B-3180	- Boilers and Related Equipment, Packaging of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-S-12514	- Starters, Motor; Across-the-Line and Reduced Voltage Types, A-C Induction, 3/4 to 100 HP, Manual and Electric.
MIL-V-13612	- Valves, Relief, Pressure and Temperature (for Hot Water Supply Systems).
MIL-B-18796	- Burners: Oil, Gas, and Gas-Oil Combination (720,000 B.T.U./Hr and Higher Input Capacity).
MIL-B-18797	- Burners Oil, Mechanical-Draft, Automatic (Under 720,000 B.T.U./Hr Input Capacity).
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.

STANDARDS

FEDERAL

FED. STD. No. 356	- Commercial Packaging of Supplies and Equipment.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IV, Low-Pressure Heating Boilers.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG1	- Motors and Generators
Publication No. ICS	- Standards for Industrial Control.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

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UNDERWRITERS LABORATORIES, INC.

UL-296 - Oil Burners.
UL-508 - Industrial Control Equipment.

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

AIR MOVING AND CONDITIONING ASSOCIATION (AMCA)

Standard 210 - Test Code for Air Moving Devices.

(Application for copies should be addressed to the Air Moving and Conditioning Association, 205 East Touhy Avenue, Park Ridge, IL 60068.)

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

COUNTY OF LOS ANGELES - AIR POLLUTION CONTROL DISTRICT

Rule 66 - Organic Solvents.

(Application for copies should be addressed to the County of Los Angeles - Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013.)

3. REQUIREMENTS

3.1 Description. The water heaters shall be furnished complete and shall include the water-jacketed combustion chamber, fuel-burning equipment, controls, safety-relief valves, and stack and smoke jack when specified and all accessories, necessary for automatic operation. The specified output shall be developed with a stack height of not more than 15 feet above the floor at sea level.

3.2 First article (preproduction model). When specified (see 6.2), the contractor shall furnish a heater for examination and test within the time frame specified (see 6.2) to prove, prior to starting production,

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that his production methods and choice of design detail will produce heaters that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Sheet steel. Sheet steel for the smokejack and stack shall be flat, zinc-coated sheet conforming to QQ-S-775, Type I, Class b.

3.3.2 Castings. Castings shall conform to QQ-I-652 (ASTM A48).

3.4 Safety. All indicating devices (such as pressure gages and thermometers) shall be marked to indicate safe and unsafe operational levels. An unpainted grounding lug or stud shall be provided, and identified as such, on all heaters. Moving parts and parts subject to high temperatures or of such nature or located so as to become a hazard to operating personnel shall be insulated, inclosed, or guarded.

3.5 Draft and output ratings. The water heaters shall deliver the output specified in Table I.

TABLE I. Output ratings

Size	Minimum Output, Btu/hr	
	On 60 Hertz Elec Power	On 50 Hertz Elec Power
600	250,000	175,000
2,000	830,000	580,000

3.5.1 Forced draft. Forced draft shall be obtained by the use of either directly or indirectly connected, electric-motor-driven fans. The fans shall be designed, constructed, and rated in accordance with AMCA Standard 210.

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3.5.2 Fuel. The water heaters shall produce the output ratings specified using fuel oil conforming to VV-F-800, Grades DF-A, or DF-2, or to VV-F-815, Grade FS, No. 1 or 2. Each burner shall be equipped with flexible tubing and fittings for making connection to an external 3/4-inch IPS fuel line 2 feet from the burner.

3.6 Construction. The water heaters shall be of the water-tube or fire-tube type of welded construction and in accordance with ASME Code, Section IV Low-Pressure Heating Boilers. Working pressure shall be 100 psig unless otherwise specified (see 6.2). Inlet and outlet tappings shall be 2-inch diameter or shall be provided with flanged, malleable iron fittings for connection to 2-inch pipe. The combustion chamber shall be of the correct proportions to produce the specified output with efficient, smokeless combustion.

3.7 Electrical system. The electrical system shall be rated for 115-volt, 60-cycle, single-phase service but shall operate as specified in 3.5 on 50-cycle power. Electrical components and wiring shall conform to UL-508 and NFPA No. 70.

3.7.1 Motors. Motors shall conform to NEMA MV-1, dripproof, splashproof, or totally enclosed, Class A insulation, and continuous duty classification based on 50°C ambient temperature of reference. When the driven equipment is operating at rated output, the load imposed on the motors shall not exceed the full-rated continuous load rating of the motors.

3.7.1.1 Starters. Starters for motors with greater than 3/4-hp rating shall conform to MIL-S-12514, Type I, Class 2, with a NEMA Type I, general-purpose enclosure.

3.7.2 Wiring. Wiring shall consist of rubber-insulated or thermoplastic-insulated wire conforming to J-C-30. Wiring shall be installed in either zinc-coated steel conduit conforming to WW-C-581, flexible steel conduit conforming to WW-C-566, or metallic tubing. Fittings shall conform to W-F-406. A plate containing a wiring diagram shall be mounted in a suitable position on the heater. The wiring diagram shall show the physical location of all wiring interconnections in the same relationship in which they are installed. Identification markings of wires, instruments, control devices, circuit-breakers, fuses, and connectors shall be shown.

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3.8 Relief valves, burner, and controls.

3.8.1 Relief valves. Relief valves shall relieve pressures in excess of 100 psi and temperatures in excess of 210° F. Temperature-relief valves shall conform to MIL-V-13612, Type II and pressure-relief valves to MIL-V-13612, Type III. The outlet(s) of the relief valve(s) shall be equipped with a discharge pipe which will direct outflow toward floor and away from operating personnel. The pipe(s) shall be of size not smaller than valve outlet(s).

3.8.2 Burner and controls. The burner and controls shall conform to UL-296. The burner shall be a single-nozzle type. The burner for the Size 600 heater shall be pressure-atomizing, gun-type and conform to MIL-B-18797, Type I, Design A. The burner for the Size 2,000 heater shall conform to MIL-B-18796, Type I, Style I, Class 2, Size 1. If one nozzle cannot provide combustion of all of the fuels specified in 3.5.2, additional replacement nozzles to provide the required range of characteristics shall be furnished with the heater (see 6.4) and the instruction plates (see 3.14) shall provide the information necessary for their selection, installation, and removal. Ignition shall be direct electric type. The heater shall be equipped with a thermostat for fully automatic operation. The thermostat shall be adjustable to control the outlet water temperature at any level between 100° F and 200° F, with a temperature-differential not more than 10° F. A water-overheat control shall also be provided which will stop operation of the burner when outlet water temperature reaches 205° F, and make it necessary to reset the control manually to restore burner operation.

3.9 Stack and smokejack. When specified (see 6.2), 12 sections of flue pipe and a smokejack shall be furnished with each heater. The flue pipe shall be of a standard commercial diameter recommended by the heater manufacturer to produce optimum draft, and shall be constructed of zinc-coated sheet steel weighing approximately 0.781 pound per square foot and conforming to QQ-S-775, Type I, Class B. A base section that adapts the smoke outlet of the heater to the flue pipe shall be furnished. The flue-pipe sections shall be shipped nested but shall be crimped on one end and provided with snaplock seams. Each section of the flue pipe shall be 24 inches in length. Seventy-two self-tapping, sheet-metal screws shall be furnished with each heater for assembly of the stack. The smokejack shall be in accordance with Figure 1, except that the components of the smokejack shall be shipped unassembled. Rain-hood supports shall be of sufficient length to locate the rain hood not less than two-thirds of one pipe diameter above the stack.

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3.10 Electromagnetic interference. When specified the electromagnetic interference emission and susceptability characteristics of the heater shall conform to MIL-STD-461, Class II B.

3.11 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.11.1 Lubricants. Lubrication, including lubrication of sealed bearings, shall be with greases conforming to MIL-G-23827 or MIL-G-10924 as follows:

(a) MIL-G-23827 - For instruments, sealed bearings and other bearings where operating temperatures are likely to exceed 175° F but do not exceed 250° F.
(b) MIL-G-10924 - For other bearings where operating temperatures do not exceed 175° F.

3.11.2 Delivery lubrication service. When delivered, all components except those for which preservatives are specified in Section 5 shall be lubricated.

3.12 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.12.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.12.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the

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contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 66, Air Pollution Control District, County of Los Angeles, CA.

3.13 Identification marking. The heater shall be identified in accordance with the manufacturer's standard commercial practice.

3.13.1 Marking. When specified (see 6.2), the water heater shall be identified in accordance with MIL-STD-130. The marking shall be applied to the water heater on identification plates conforming to MIL-P-514, Type I, Style 3, Composition C of Type I, Grade A, Class 1 material.

3.14 Instruction plates. Each water heater shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the water heater. The plates shall be of the same type as used by the contractor in his normal commercial practice.

3.14.1 Plates. When specified, the instruction plates shall conform to MIL-P-514, Type III, size optional, Composition C of Type I, Grade A, Class I material. Plates shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.15 Treatment and painting. The heater shall be finished and painted in accordance with the manufacturer's standard practice for this type of equipment.

3.16 Repair parts. When specified, repair parts shall be furnished with each heater in the quantity specified and shall be packed and shipped with the heater (see 6.2 and 6.4).

3.17 Repair parts. When specified, repair parts shall be furnished with each heater in the quantity specified and shall be packed and shipped with the heater (see 6.2 and 6.4).

3.17 Instructional material. Instructions furnished with each heater shall be the contractor's standard commercial material. The material shall include installation instructions for such items which are shipped detached from the heater proper.

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3.18 Workmanship. The general construction and assembly of the water heater shall be of a neat and workmanlike character with parts well fitted and bolts and other fasteners drawn up tightly to give rigidity. Any exposed edges which might reasonably be brought in contact with the hand in normal adjustment or usage shall be smooth. All parts, components, and assemblies of the water heater including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful, extraneous material. All edges shall be rounded or beveled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The water heaters shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. When specified (see 6.2), the water heater shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

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4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, Inspection Level S-1.

4.4.2 Lot. A lot shall consist of all heaters of each size offered for delivery at the same time under a specific contract.

4.4.3 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective.

4.4.4 Tests.

4.4.4.1 Sampling. Sampling for tests shall consist of one heater of each size from each lot which shall be subjected to the tests specified in 4.5.2.3 and 4.5.2.4. Failure of the sample to pass any one of the tests specified shall be cause for rejection of the whole lot.

4.5 Inspection procedure.

4.5.1 Examination. The water heaters shall be examined for the following defects:

101. Absence of ASME stamps on boiler and relief valves, UL label on oil burner, or nonconformance to referenced publications.
102. Material not as specified.
103. Safety devices missing or inadequate.
104. Electrical wiring and components not as specified.
105. Relief valves, burners, and controls not as specified.
106. Dimensions not as specified.
107. Parts or components missing, do not function, or not assembled as specified.
108. Components not accessible for maintenance.
109. Screws and other fastenings for flue-pipe and smokejack do not fit holes in flue-pipes or other parts.
110. Workmanship not as specified.
111. Identification marking not as specified.

4.5.2 Tests. Tests shall be performed as follows.

4.5.2.1 Output. Equip the water heater with a positive-action-type flowmeter and a 0° F to 100° F thermometer in the water inlet line and a 0° F to 200° F thermometer in the water outlet line. Start and operate

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the heater on 60 cycle power at maximum output (see 3.5) with a water-temperature differential of 50° F to 55° F between the inlet and the outlet. The inlet water temperature shall be the normal temperature prevailing in the water supply used, provided temperature fluctuatious do not exceed 3° F over the 1 hour test period. Record the water flow and temperature differential for 1-hour period. Repeat the test, starting and operating on 50 cycle power. Compute the heater output using the following equation:

$$H = G(t2 - t1)\ 500$$

where H = output in Btu/hr
G = rate of water flow in gallons per minute
t2 = leaving water temperature in degrees fahrenheit
t1 = entering water temperature in degrees fahrenheit.
500 = constant of computation.

A Btu output of less than that specified in Table I (see 3.5) shall constitute failure of this test.

4.5.2.2 Temperature-control. Reconnect to 60 cycle power. Set the water temperature control at 150° F (see 3.8.2). Increase the inlet water temperature or reduce the water flow to obtain outlet water temperature in excess of 150° F. Measure and record for a period of not less than 1 hour and not less than 10 cycles of the burner, the maximum and minimum outlet water temperatures. A differential in excess of 10° F shall constitute failure of this test.

4.5.2.3 Pressure. Render inoperative or remove the pressure relief valve(s), plug opening(s), and apply a hydrostatic pressure of 150 psig or 1.5 times working pressure if latter is different from 100 psig. Any deformation, leakage, or rupture of the water heater shall constitute failure of this test (see 3.6).

4.5.2.4 Relief valve.

4.5.2.4.1 Pressure relief. Subject the water heater to a pressure necessary to open the relief valve(s). Nonopening of the valve(s) when the pressure exceeds 100 psig (or other working pressure specified) shall constitute failure of this test.

4.5.2.4.2 Temperature relief. Rig to prevent operation of the water-temperature thermostats. Raise the temperature in the water heater to a temperature necessary to operate the temperature-relief valves. Nonoperation of the temperature-relief valve when the temperature exceeds 210° F shall constitute failure of this test.

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4.5.2.5 Flame failure. Test for flame failure due to combustion air failure, interruption of atomization (if applicable) and power interruption in accordance with UL-296, Test Nos. 3, 4, and 6 respectively. Inability to shut off fuel to the burner within the safe time period specified in UL-296 and to require manual restart, or to accomplish a safe automatic restart upon restoration to normal conditions shall constitute failure of this test.

4.5.2.6 Water overheat. Rig to prevent operation of the regular water-temperature-control thermostat. Raise the temperature of the outlet water until the water-overheat control stops operation of the burner. Inability to stop operation when the outlet temperature exceeds 205° F or restoration of burner operation without manual resetting shall constitute failure of this test.

4.5.2.7 Electromagnetic interference. When specified (see 3.10 and 6.6), the preproduction model shall be equipped for the reduction of electromagnetic interference and be tested in accordance with MIL-STD-461. The contractor shall furnish the contracting officer the test report required by MIL-STD-461 within the time frame specified (see 6.2). Upon approval of the report by the contracting officer and provided all other requirements of the specification are met, the preproduction model shall be used as a model for all production water heaters. Disapproval of the report shall constitute failure of this test.

4.6 Inspection of packaging. The preservation and packing shall be examined and tested to determine compliance with the Quality Assurance Provisions of MIL-B-3180. The marking for commercial shipment shall be examined as specified for Level A, B, or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or B, or Commercial as specified (see 6.2).

5.1.1 Level A. Each complete heater shall be preserved in accordance with the Level A preservation and packaging requirements of MIL-B-3180.

5.1.2 Level B. Each complete heater shall be preserved in accordance with the Level B preservation and packaging requirements of MIL-B-3180.

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5.1.3 Commercial. Each complete heater shall be preserved in accordance with FED. STD. No. 356.

5.2 Packing. Packing shall be Level A or B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Heaters, preserved as specified in 5.1, shall be packed in accordance with the Level A packing requirements of MIL-B-3180.

5.2.2 Level B. Heaters, preserved as specified in 5.1, shall be packed in accordance with the Level B packing requirements of MIL-B-3180.

5.2.3 Commercial. Heaters, preserved as specified in 5.1, shall be packed in accordance with FED. STD. No. 356.

5.3 Marking.

5.3.1 Military packaging. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking for commercial packaging shall be in accordance with FED. STD. No. 356.

6. NOTES

6.1 Intended use. The water heaters are intended for use as sources of hot water in theaters of operation and in temporary buildings generally.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of heater required (see 1.2).
(c) When preproduction model is required and time frame involved (see 3.2, 4.3.2, 6.3 and 6.5).
(d) If working pressure other than 100 psig is required (see 3.6 and 4.5.2.3).
(e) If stack and smokejack are required (see 3.9 and 6.4).
(f) When special marking is required (see 3.13.1).
(g) When special instruction plates are required (see 3.14.1).
(h) Repair parts required (see 3.16).

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(i) When quality conformance tests are required (see 4.4.4.1 and 6.5).
(j) Whether EMI provisions are required and time frame for submittal of request (see 3.10, 4.5.2.7 and 6.6).
(k) Degree of preservation and packing required (see 5.1 and 5.2).

6.3 Preproduction model. Any changes or deviations of production heaters from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish heaters conforming to this specification.

6.4 Items to be furnished with water heater. No shipment of water heaters shall be made unless repair parts, replacement nozzles if required (see 3.8.2), maintenance tools, operating tools, and accessories are included with the shipment or unless approval for shipment of the water heater without such items has been received from the contracting officer.

6.5 Test information. If contractor has successfully manufactured the heaters over the past 5 years, or if delivery to the Government of the same item has occurred within the past 12 months, the preproduction tests should be waived. If preproduction tests are required the quality conformance tests should be waived. Quotations should be requested both with preproduction tests and without.

6.6 Electromagnetic interference. This requirement shall only be included when the heater is required to operate in tactical areas. The contracting officer should forward a copy of the test report (see 4.5.2.7) to the Commanding General, U.S. Army Electronics Command ATTN: AMSEL-RD-GF, Fort Monmouth, New Jersey 07703.

Custodians:
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Air Force - 84

Review activity:
DSA - CS

User activities:
Army - AV, GL, CE

Preparing activity:
Army - ME

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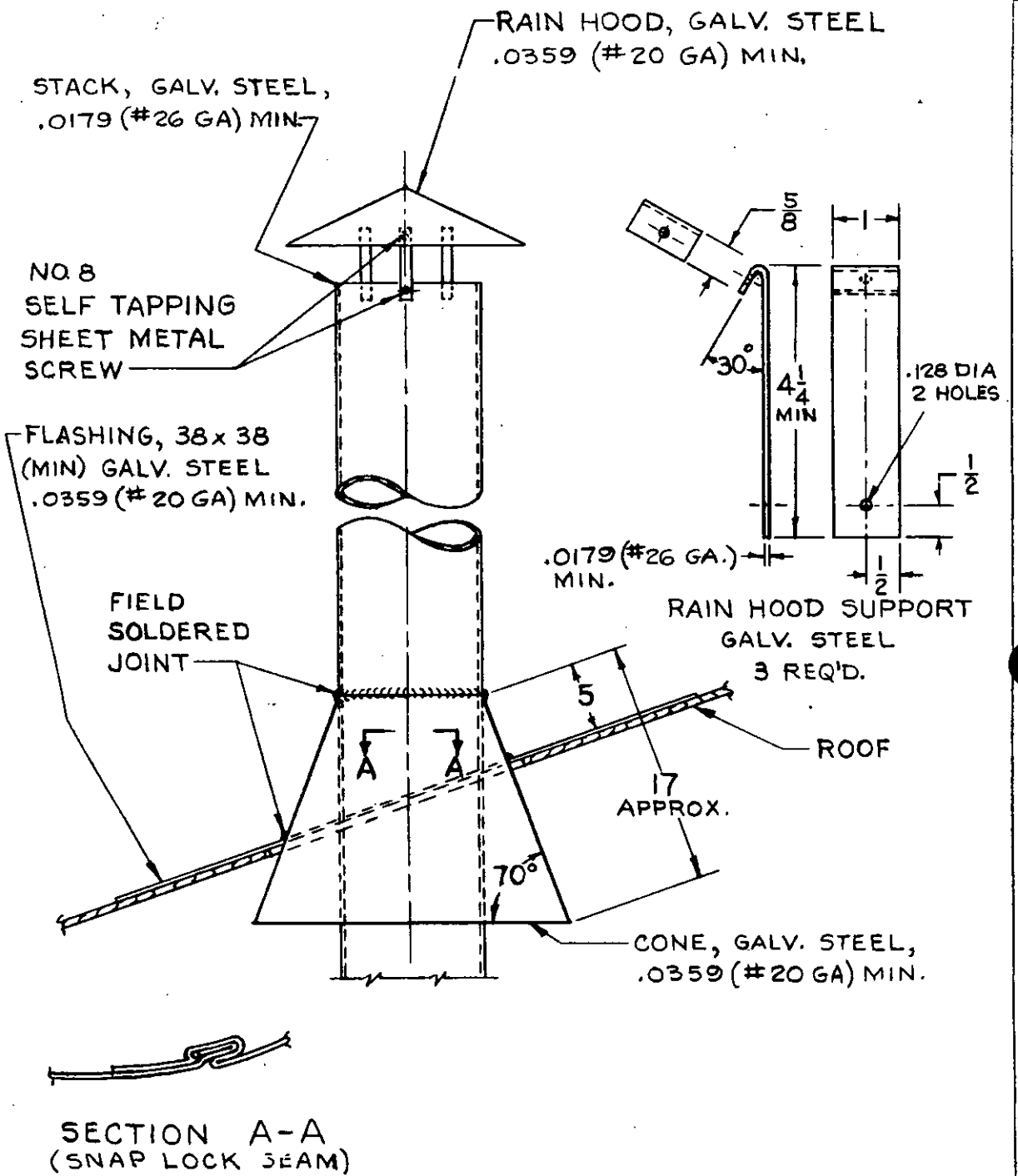


FIG. 1 STACK AND SMOKE JACK

X-3152

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-H-12394D Heaters, Water, Liquid Fuel

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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